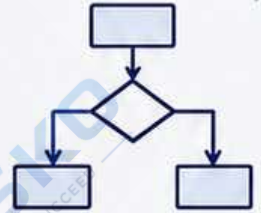


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EXAM SAMPLE PAPER

-  Section A – Short Answer Questions
(Attempt all)
-  Section B – Long Answer Questions
(Attempt any two)
-  Section C – Short Answer Questions
(Attempt all)
-  Section D – Long Answer Questions
(Attempt any two)
-  Section E – Short Answer Questions
(Attempt all)

Note: Question pattern may vary as
per university guidelines.



Semester V – Core Java

Sample Paper – 1 (Pattern A)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

Q1. Write answers in one or two lines and solve **any six**.

- i) What is Java?
- ii) What is JVM?
- iii) Write the full form of JDK.
- iv) What is a keyword in Java?
- v) Define the **if-else** statement.
- vi) What is the use of a **for** loop?
- vii) What is an object?
- viii) What is inheritance?

Q2. Answer in short **any two** of the following(3 marks each.)

- A) Explain any three features of Java.
- B) Differentiate between JDK, JRE and JVM.
- C) Explain different data types in Java.
- D) Explain the structure of a Java program with a simple example.

Q3. Explain in detail **any one** of the following(6 marks each.)

E) Explain different decision-making statements in Java: **if**, **if-else**, nested **if-else**, **else-if** ladder and **switch**, with suitable examples.

F) Explain looping statements in Java. Discuss **while**, **do-while** and **for** loops with suitable examples. Also explain jump statements used in loops.

Q4. Answer in short **any two** of the following(3 marks each.)

A) Define Class and Object. Explain their relationship.

B) What is Method Overloading? Give a suitable example.

C) Explain Constructor and Constructor Overloading.

D) Explain the **public**, **private** and **protected** access specifiers.

Q5. Explain in detail **any one** of the following(6 marks each.)

E) Define Inheritance. Explain Super Class and Extended Class with a suitable Java example.

F) Explain Interfaces in Java. Describe defining, extending and implementing an interface with a suitable example.

Sample Paper – 2 (Pattern B)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

Q1. Write answers in one or two lines and solve **any six**.

- i) Who developed Java?
- ii) What is JRE?
- iii) What is a command line argument?
- iv) What is a **switch** statement?
- v) Define constructor.
- vi) What is the **this** keyword?
- vii) What is an abstract class?
- viii) What is an interface?

Q2. Explain in detail **any one** of the following(6 marks each.)

E) Explain the history and evolution of Java. Discuss its important features and explain the role of JDK, JRE and JVM.

F) Explain Java data types, keywords and operators. Write a simple Java program and explain its basic structure.

Q3. Answer in short **any two** of the following(3 marks each.)

- A) Explain the **else-if** ladder with a suitable example.
- B) Differentiate between **while** and **do-while** statements.
- C) Explain the **switch** statement in Java.
- D) What are jump statements in loops? Explain with suitable examples.

Q4. Explain in detail **any one** of the following(6 marks each.)

E) Explain classes and objects in Java. Describe defining a class, declaration of fields, methods, object creation and accessing class members with a suitable example.

F) Explain constructors and constructor overloading. Also discuss method overloading and the use of the **new** and **this** keywords.

Q5. Answer in short **any two** of the following(3 marks each.)

A) What is inheritance? Explain the concept of Super Class and Extended Class.

B) Explain the **final** variable, **final** method and **final** class.

C) What is an abstract method? Explain abstract class.

D) Explain the difference between extending and implementing an interface.

Semester V – Software Engineering

Sample Paper – 1 (Pattern A)

Time: 2 Hours

Maximum Marks: 30

Instructions:

- 1. Question No. 1 is compulsory.**
- 2. Draw neat, well-labelled diagrams wherever necessary.**
- 3. Students may use various colours to signify answers.**

Q1. Write answers in one or two lines and solve **any six**.

- Define Software Engineering.
- What is SDLC?
- What is a Prototype Model?
- Define Functional Requirement.
- What is Risk Identification?
- What is Software Architecture?
- Define Black-Box Testing.
- What is Debugging?

Q2. Answer in short **any two** of the following(3 marks each.)

- Explain the characteristics of software.
- Why is Software Engineering needed?
- Explain the phases of SDLC.
- Write a short note on the Waterfall Model.

Q3. Explain in detail **any one** of the following(6 marks each.)

E) Explain Functional and Non-functional Requirements. Also discuss User Requirements and System Requirements.

F) Explain the Requirements Engineering Process in detail, including Requirements Elicitation and Analysis, Requirements Validation and Requirements Management.

Q4. Answer in short **any two** of the following(3 marks each.)

A) What is Software Risk? Explain any two types of software risks.

B) Explain Risk Identification.

C) What is the RMMM Plan?

D) Explain the Golden Rules of User Interface Design.

Q5. Explain in detail **any one** of the following(6 marks each.)

E) Explain the strategic approach to software testing. Discuss test strategies for conventional software.

F) Explain Black-Box Testing and White-Box Testing. Differentiate between them with suitable examples.

Sample Paper – 2 (Pattern B)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

Q1. Write answers in one or two lines and solve **any six**.

- i) What is a Process Model?
- ii) Define Spiral Model.
- iii) What is a System Requirement?
- iv) What is Requirements Validation?
- v) Define RMMM Plan.
- vi) What is an Architectural Pattern?
- vii) What is Validation Testing?
- viii) Define System Testing.

Q2. Explain in detail **any one** of the following(6 marks each.)

E) Explain Software Development Life Cycle (SDLC). Describe all the phases of SDLC in detail.

F) Explain Software Development Models. Discuss Waterfall Model, Prototype Model and Spiral Model with their characteristics.

Q3. Answer in short **any two** of the following(3 marks each.)

- A) Differentiate between Functional and Non-functional Requirements.
- B) Explain User Requirements and System Requirements.
- C) What is Requirements Elicitation and Analysis?
- D) Explain Requirements Management.

Q4. Explain in detail **any one** of the following(6 marks each.)

E) Explain Risk Management in Software Engineering. Discuss Risk Strategies, Software Risks, Risk Identification and the RMMM Plan.

F) Explain Design Engineering and Software Architecture. Discuss the Design Process, Design Quality, Data Design and Architectural Patterns.

Q5. Answer in short **any two** of the following(3 marks each.)

A) What is the strategic approach to software testing?

B) Differentiate between Black-Box Testing and White-Box Testing.

C) What is Validation Testing?

D) Explain the role of debugging in software testing.

Semester V – Advance Python

Sample Paper – 1 (Pattern A)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. **Question No. 1 is compulsory.**
2. **Draw neat, well-labelled diagrams wherever necessary.**
3. **Students may use various colours to signify answers.**

Q1. Write answers in one or two lines and solve **any six**.

- i) What is string slicing in Python?
- ii) What is a List?
- iii) Define Dictionary.
- iv) What is a Tuple?
- v) What is a Set?
- vi) What is the Pickle Module?
- vii) What is a Regular Expression?
- viii) Define Encapsulation.

Q2. Answer in short **any two** of the following(3 marks each.)

- A) Explain string indexing and slicing with suitable examples.
- B) Explain any three commonly used string methods.
- C) Explain list indexing and slicing with suitable examples.
- D) Explain how key:value pairs are accessed and modified in a Dictionary.

Q3. Explain in detail **any one** of the following(6 marks each.)

E) Explain Tuples in Python. Discuss creation, basic operations, indexing, slicing, built-in functions and Tuple methods with suitable examples.

F) Explain Sets in Python. Discuss set methods, traversing of sets, `zip()` function and Frozen Set with suitable examples.

Q4. Answer in short **any two** of the following(3 marks each.)

A) Explain the different types of files in Python.

B) Write a short note on the Pickle Module.

C) What is CSV? Explain reading and writing CSV files.

D) Explain special characters used in Regular Expressions.

Q5. Explain in detail **any one** of the following(6 marks each.)

E) Explain Classes and Objects in Python. Describe class creation, object creation, constructor method and classes with multiple objects.

F) Explain Encapsulation, Inheritance and Polymorphism in Python with suitable examples.

Sample Paper – 2 (Pattern B)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

Q1. Write answers in one or two lines and solve **any six**.

- i) What is string formatting in Python?
- ii) What is a List Method?
- iii) Define a Tuple.
- iv) What is a Frozen Set?
- v) What is the `os.path` module?
- vi) Define Named Groups in Regular Expressions.
- vii) What is a Constructor in Python?
- viii) What is Polymorphism?

Q2. Explain in detail **any one** of the following(6 marks each.)

E) Explain Strings in Python. Discuss basic string operations, indexing, slicing, string formatting and commonly used string methods with suitable examples.

F) Explain Lists and Dictionaries in Python. Discuss list operations, built-in functions, list methods and accessing and modifying key:value pairs in dictionaries.

Q3. Answer in short **any two** of the following(3 marks each.)

- A) Explain the relation between Tuples and Lists.
- B) Explain any three built-in functions used with Tuples.
- C) What is the `zip()` function? Explain its use.
- D) Differentiate between a Set and a Frozen Set.

Q4. Explain in detail **any one** of the following(6 marks each.)

E) Explain file handling in Python. Discuss creating and reading text files, file methods for reading and writing data, and reading and writing binary files.

F) Explain Regular Expression operations in Python. Discuss special characters, Regular Expression methods and Named Groups with suitable examples.

Q5. Answer in short **any two** of the following(3 marks each.)

A) Differentiate between Class Attributes and Data Attributes.

B) Explain the Constructor Method in Python.

C) What is Inheritance?

D) Explain the concept of Multiple Objects in a Class.

Semester V – Data Science

Sample Paper – 1 (Pattern A)

Time: 2 Hours

Maximum Marks: 30

Instructions:

- 1. Question No. 1 is compulsory.**
- 2. Draw neat, well-labelled diagrams wherever necessary.**
- 3. Students may use various colours to signify answers.**

Q1. Write answers in one or two lines and solve **any six**.

- Define Data Science.
- What is the Data Science Lifecycle?
- What is structured data?
- Define data cleaning.
- What is a histogram?
- Define descriptive statistics.
- What is supervised learning?
- Define clustering.

Q2. Answer in short **any two** of the following(3 marks each.)

- Explain the importance of Data Science.
- Describe the Data Science Lifecycle.
- Explain different types of data.
- Write a short note on applications of Data Science.

Q3. Explain in detail **any one** of the following(6 marks each.)

E) Explain different data collection methods. Discuss structured and unstructured data with suitable examples.

F) Explain data preprocessing in detail. Discuss data cleaning, handling missing values, data transformation and normalization.

Q4. Answer in short **any two** of the following(3 marks each.)

A) Explain the importance of data visualization.

B) Differentiate between a Bar Chart and a Pie Chart.

C) What is a Histogram? Explain its purpose.

D) Define Mean, Median and Mode.

Q5. Explain in detail **any one** of the following(6 marks each.)

E) Explain Machine Learning and its major types. Discuss Supervised and Unsupervised Learning with suitable examples.

F) Explain Classification, Regression and Clustering. Also discuss their applications in real life.

Sample Paper – 2 (Pattern B)

Time: 2 Hours

Maximum Marks: 30

Instructions:

1. **Question No. 1 is compulsory.**
2. **Draw neat, well-labelled diagrams wherever necessary.**
3. **Students may use various colours to signify answers.**

Q1. Write answers in one or two lines and solve **any six**.

- i) What is Big Data?
- ii) What is meant by Data Science tools and platforms?
- iii) Define data transformation.
- iv) What is data normalization?
- v) What is a scatter plot?
- vi) Define correlation.
- vii) What is unsupervised learning?
- viii) What is classification?

Q2. Explain in detail **any one** of the following(6 marks each.)

E) Define Data Science and explain its importance. Describe the Data Science lifecycle and its major stages.

F) Explain the different types of data and discuss the applications of Data Science. Also explain the role of Big Data and AI in Data Science.

Q3. Answer in short **any two** of the following(3 marks each.)

- A) Explain structured and unstructured data with examples.
- B) What are the different methods of data collection?
- C) Explain handling of missing values in a dataset.
- D) Write a short note on data integration and data reduction.

Q4. Explain in detail **any one** of the following(6 marks each.)

E) Explain different types of charts and graphs used in Data Science. Discuss Bar Chart, Pie Chart, Histogram and Scatter Plot with their uses.

F) Explain descriptive statistics. Discuss Mean, Median, Mode, Correlation and Regression basics with suitable examples.

Q5. Answer in short **any two** of the following(3 marks each.)

A) Differentiate between supervised and unsupervised learning.

B) Explain regression in Machine Learning.

C) What is clustering? Give a suitable example.

D) Write a short note on ethical issues in Machine Learning.